Hugo Jarkoff

Senior Machine Learning Engineer

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5+ years designing, deploying, and scaling production ML systems across computer vision, generative AI, and cloud infrastructure.

Experience

- **Addactis (formerly NamR, acquired in 2025)** — Paris, France
  Senior ML/CV Software Engineer — 2024 - present

  Responsible for production ML systems powering aerial imagery processing and SaaS products:
  - Refactored the team's ML training framework, reducing time from data acquisition to production release by over 50%.
  - Redesigned the aerial imagery preprocessing pipeline with serverless Cloud Run APIs, cutting legacy codebase size down and reducing data integration time for ML workflows.
  - Refactored the aerial imagery inference pipeline into a distributed GPU batch-processing system with PostgreSQL-coordinated workers, achieving an 80% faster runtime and 90% lower compute cost on large-scale datasets.
  - Designed, built, and deployed an AI agent platform (FastAPI + Pydantic-AI) integrated into production SaaS insurance products.

- **Finegrain** — Paris, France
  ML/CV Software Engineer — 2024 (4 months)

  State-of-the-art generative model fine-tuning for image editing:
  - Fine-tuned foundation models (SAM, Stable Diffusion) using adapters (LoRA, IP-Adapter) for image editing tasks.
  - Conducted continuous literature review in generative modeling.
  - Contributed to the team's high-performance OSS micro-framework for foundation model adaptation (Refiners).

- **Invoxia** — Paris, France
  R&D Machine Learning Engineer — 2021 – 2024

  Led multiple end-to-end ML projects from research to deployment:
  - Designed a CV-based signal processing model to estimate canine respiratory rate from accelerometer data, achieving 98.5% accuracy (preprint available).
  - Researched and adapted neural architectures (CNNs, Vision Transformers), focusing on model compression for embedded deployment.
  - Contributed to the inference system for trained models, serving 10k+ devices via batch processing through Kubernetes and AWS SageMaker, with real-time monitoring platform (Grafana, Prometheus).

Education

- **ISAE-SUPAERO** — Toulouse, France
  Engineering Degree (MSc.) — 2016 – 2020
  - *Major in Machine Learning*: Deep Learning (Computer Vision, NLP), RL, MLOps.
  - *Minor in Advanced Mathematics*: Bayesian statistics, HPC, optimization.
  - *Final year Team Project for Airbus*: Developed an airport image segmentation and classification system using fully convolutional neural networks (UNet).

- **Lycée Louis-le-Grand** — Paris, France
  Preparatory Classes — 2014 – 2016
  - *PCSI - PC**: Mathematics - Physics - Chemistry - Computer Science.

Technical Skills

- **Programming**: advanced knowledge of Python (multithreaded programming, async programming); intermediate knowledge of C++ / C / Bash.
- **Deep Learning**: PyTorch, TensorFlow, Keras.
- **MLOps & Infrastructure**: Docker, Kubernetes, Terraform, NVIDIA Triton, PyTriton, CI/CD (GitLab), experiment tracking (WandB, ClearML).
- **Databases**: PostgreSQL, PostGIS.
- **Cloud Platforms**: Google Cloud Platform (GCP), Amazon Web Services (AWS).

Languages

- **French**: Native language.
- **English**: Full professional proficiency.
  TOEFL ITP: 633/677.

Personal Projects and Diverse Interests

- **Language Models Training**: RapGPT: Trained a Transformer-based language model from scratch (architecture, training loop, and deployment) to generate French rap lyrics; deployed live demos on Hugging Face (with open weights).
- **Systems Programming**: Developed a simple, lightweight (700KB running), multiprocess (fork) HTTP server in C++.
- **Rock Climbing**: Experienced climber (20+ years of practice). Designed and built a custom finger-force measurement device (ESP32 soldered with deformation gauge, embedded with C software).